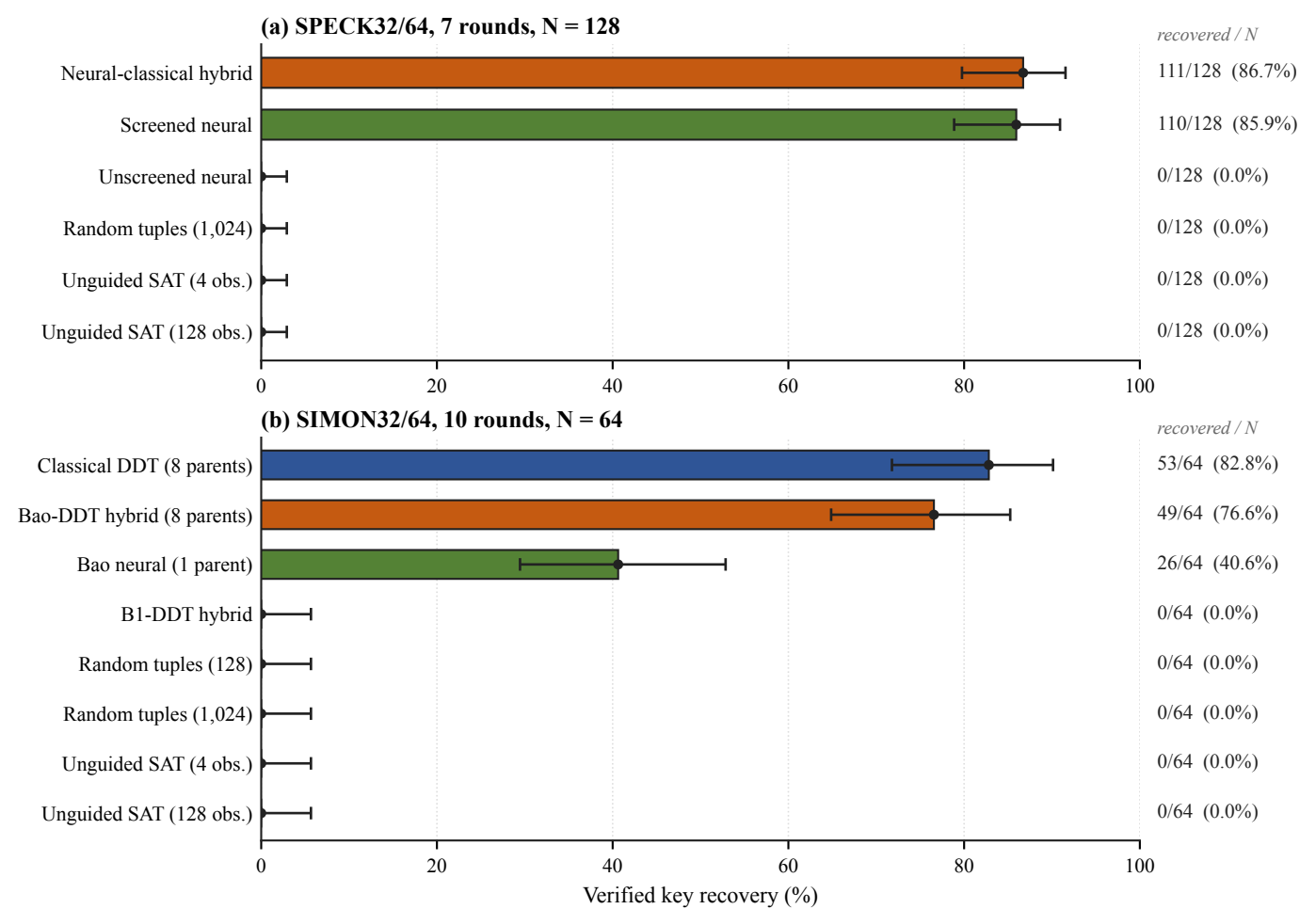

(a) SPECK32/64, 7 rounds, N = 128
recovered / N
Neural-classical hybrid
111/128 (86.7%)
Screened neural
110/128 (85.9%)
Unscreened neural
0/128 (0.0%)
Random tuples (1,024)
0/128 (0.0%)
Unguided SAT (4 obs.)
0/128 (0.0%)
Unguided SAT (128 obs.)
0/128 (0.0%)
0
20
40
60
80
100
(b) SIMON32/64, 10 rounds, N = 64
recovered / N
Classical DDT (8 parents)
53/64 (82.8%)
Bao-DDT hybrid (8 parents)
49/64 (76.6%)
Bao neural (1 parent)
26/64 (40.6%)
B1-DDT hybrid
0/64 (0.0%)
Random tuples (128)
0/64 (0.0%)
Random tuples (1,024)
0/64 (0.0%)
Unguided SAT (4 obs.)
0/64 (0.0%)
Unguided SAT (128 obs.)
0/64 (0.0%)
0
20
40
60
80
100
Verified key recovery (%)